

PROJECT TITAN

Educational Risk Analysis (FMEA)

Educational Engineering Project

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IMPORTANT SCOPE NOTICE

Project TITAN is an educational design and analysis project. It is not a clinically validated device, regulatory submission, or product cleared for human use.

Document Control and Revision History

Revision	Date	Description	Prepared By	Status
A	July 2026	Initial educational design FMEA for mechanical, manufacturing, assembly, and use-related hazards.	Michael Lozada-Longog	Released

Source baseline: Project TITAN Engineering Design Report (July 2026), CAD assembly, completed engineering drawings, and Fusion 360 static FEA results documented by the author.

1. Purpose and Method

This document applies a simplified Failure Modes and Effects Analysis to Project TITAN. It is intended to demonstrate risk-based engineering thinking. It is not an ISO 14971-compliant risk-management file, clinical hazard analysis, or regulatory submission.

Scales: Severity (S), Occurrence (O), and Detection (D) are rated 1–10. Risk Priority Number (RPN) = $S \times O \times D$. Ratings are preliminary engineering estimates and must be updated with prototype and test data.

2. Rating Guidance

Rating	Severity	Occurrence	Detection
1–2	Negligible effect	Remote	Almost certain detection
3–4	Minor functional degradation	Low	High likelihood of detection
5–6	Loss of intended function or repair required	Moderate	Moderate likelihood of detection
7–8	Major structural/function loss; potential injury in a future wearable context	High	Low likelihood of detection
9–10	Catastrophic failure or severe injury potential	Very high	Unlikely to be detected before effect

3. Preliminary Design FMEA

ID	Failure Mode	Cause	Effect	S	O	D	RPN	Recommended Control
FM-01	Bearing seizure or binding	Contamination, misalignment, inadequate clearance	Loss of smooth rotation; increased load into structure	7	4	5	140	Use sealed COTS bearings; define fits; verify alignment and rotational torque
FM-02	Bearing	Insufficient	Loss of	6	4	6	144	Add

ID	Failure Mode	Cause	Effect	S	O	D	RPN	Recommended Control
	loosens in plate bore	retention or incorrect fit	alignment and local wear					shoulder/retainer; specify fit and inspection method
FM-03	Side plate yields or cracks	Overload, stress concentration, cyclic fatigue	Permanent deformation or structural separation	9	3	5	135	Maintain SF target; add fatigue study; inspect high-stress regions
FM-04	Main shaft bends	Excess radial load or inadequate shaft material	Misalignment, bearing damage, loss of motion	8	3	5	120	Hand-calculate bending; reconsider steel shaft; test deflection
FM-05	Main shaft surface wears	Aluminum shaft against bearing/interface; contamination	Clearance growth and poor alignment	6	5	6	180	Use hardened steel/stainless shaft or hardened sleeve; define lubrication
FM-06	Spacer wear or deformation	Incorrect material, edge loading, poor fit	Axial play and component rubbing	5	4	5	100	Specify material and flatness; verify stack-up
FM-07	Fastener loosening	Vibration, insufficient torque, no locking feature	Joint motion and progressive damage	8	4	5	160	Define fastener grade, torque, locking method, inspection interval
FM-08	Fastener pull-through or hole elongation	Thin section, high bearing stress, repeated motion	Loss of rigid frame connection	8	3	6	144	Check bearing stress; add washers/bushings; inspect holes
FM-09	Upper/lower frame bends excessively	High moment, insufficient stiffness	Poor alignment and reduced function	7	3	5	105	Verify displacement; increase section or add ribs if needed

ID	Failure Mode	Cause	Effect	S	O	D	RPN	Recommended Control
FM-10	Interference during rotation	Inadequate clearance or assembly error	Binding, sudden stop, part damage	6	4	3	72	CAD interference review; physical motion test; clearance requirement
FM-11	Incorrect assembly order	Ambiguous instructions or mirrored parts	Misalignment, interference, incomplete engagement	5	4	3	60	Number parts; create assembly instructions and inspection checklist
FM-12	Incorrect material installed	Procurement or labeling error	Unexpected strength, corrosion, or wear behavior	7	2	6	84	Material certs; part marking; incoming inspection
FM-13	Sharp edges or burrs	Incomplete deburring	Cuts, snagging, stress risers	6	4	2	48	Deburr and edge-break requirement; visual inspection
FM-14	Corrosion or sweat exposure damage	Humidity, salts, dissimilar-metal contact	Surface degradation, seizure, reduced life	6	4	6	144	Anodize/coat; isolate dissimilar metals; environmental test
FM-15	FEA result misinterpreted	Idealized contacts, outdated mesh, incorrect constraints	Unsafe design decision based on invalid evidence	9	3	7	189	Document assumptions; mesh check; peer review; correlate with test
FM-16	Load applied asymmetrically	Incorrect use or uneven interface	Unexpected torsion and one-sided plate stress	8	4	6	192	Add off-axis load case and torsional analysis
FM-17	Range of motion exceeds safe mechanical limit	No validated hard stop in baseline	Hyperextension or component collision in future wearable concept	9	4	7	252	Design adjustable hard stop; verify angular limits before human-use consideration
FM-18	Device used on a person before	Scope misunderstanding	Potential injury	10	2	8	160	Prominent educational-use labeling;

ID	Failure Mode	Cause	Effect	S	O	D	RPN	Recommended Control
	validation							prohibit human testing without formal review

4. Risk Prioritization

Priority	ID	Failure Mode	Initial RPN	Disposition
1	FM-17	Range of motion exceeds safe mechanical limit	252	Action required before physical/wearable V2
2	FM-16	Load applied asymmetrically	192	Action required before physical/wearable V2
3	FM-15	FEA result misinterpreted	189	Action required before physical/wearable V2
4	FM-05	Main shaft surface wears	180	Action required before physical/wearable V2
5	FM-18	Device used on a person before validation	160	Track and verify
6	FM-07	Fastener loosening	160	Track and verify
7	FM-14	Corrosion or sweat exposure damage	144	Track and verify
8	FM-08	Fastener pull-through or hole elongation	144	Track and verify

5. Required Risk Controls Before Prototype

1. Revisit the main shaft material and bearing interface.
2. Define bearing retention and fit tolerances.
3. Select and document fasteners, torque, and locking methods.
4. Add an off-axis/torsional load case and preliminary fatigue analysis.
5. Create assembly, inspection, and labeling instructions.
6. Add prominent “not for human use” labeling to the prototype and technical package.
7. Plan physical static-load and deflection tests before any wearable evaluation.

6. Verification of Risk Controls

Control	Verification Method	Acceptance Evidence	Status
Minimum structural safety factor	FEA and later physical load test	FEA SF ≥ 2.0 ; test criterion TBD	FEA complete; test planned
Bearing alignment	CAD inspection and rotational torque test	No binding; torque limit TBD	CAD complete

Control	Verification Method	Acceptance Evidence	Status
Bearing retention	Drawing review and pull-out inspection	Retention design and test value TBD	Open
Fastener security	Torque spec and cyclic inspection	No loosening after defined cycles	Open
Edge safety	Visual and tactile inspection	No burrs or sharp edges	Future fabrication
Scope control	Label/document review	Educational-use notice present	Documented

7. Residual Risk Statement

Residual risk cannot be formally accepted because no physical prototype, fatigue test, misuse evaluation, or human-factors assessment has been completed. The current FMEA supports design planning only. Project TITAN must remain a non-clinical educational device until future verification evidence is generated and reviewed.

8. FMEA Maintenance

Update this FMEA whenever the geometry, material, fastener strategy, load case, bearing retention, prototype process, or intended use changes. New test data should replace preliminary occurrence and detection ratings.

References

TITAN-PRD-001, Product Requirements Document, Revision B.

TITAN-SYS-001, System Architecture Document, Revision A.

Project TITAN Engineering Design Report, July 2026.